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Improved protein A resin for antibody capture in a continuous countercurrent tangential chromatography system

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Abstract

Continuous countercurrent tangential chromatography (CCTC) enables steady-state continuous bioprocessing with low-pressure operation and high productivity. CCTC has been applied to initial capture of monoclonal antibodies (mAb) from clarified cell culture harvest and postcapture polishing of mAb; however, these studies were performed with commercial chromatography resins designed for conventional column chromatography. In this study, a small particle size prototype agarose resin (20–25 µm) with lower cross-linking was co-developed with industrial partner Purolite and tested with CCTC. Due to increased binding capacity and faster kinetics, the resulting CCTC process showed more than a 2X increase in productivity, and a 2X reduction in buffer consumption over commercial protein A resins used in previous CCTC studies, as well as more than a 10X productivity increase versus conventional column operation. Single-pass tangential flow filtration was integrated with the CCTC system, enabling simple control of eluate concentration. A scale-up exercise was conducted to provide a quantitative comparison of CCTC and batch column chromatography. These results clearly demonstrate opportunities for using otherwise unpackable soft small particle size resins with CCTC as the core of a continuous bioprocessing platform.

KEYWORDS

antibody, continuous chromatography, protein A, purification, single-use

1 | INTRODUCTION

Monoclonal antibody (mAb) based drugs are used for treating diseases, such as cancers, psoriasis, and rheumatoid arthritis and are one of the main sources of growth in the biotechnology sector. Downstream purification of monoclonal antibodies is typically performed using packed chromatography columns in batch mode, with the initial capture by protein A. Typical columns are expensive stainless steel that is not suitable for single-use facilities.

Continuous countercurrent tangential chromatography (CCTC) provides an attractive alternative to batch column chromatography (Dutta et al., 2015; Dutta, Tan, Napadensky, Zydney, & Shinkazh, 2016, 2017;

Napadensky, Shinkazh, Teella, & Zydney, 2013; Shinkazh et al., 2011). The system performs the chromatographic operations (binding, wash, elution, strip, and equilibration) sequentially, with each step composed of multiple stages consisting of hollow fiber membranes and static mixers. Unlike column chromatography, where the resin particles are packed in a fixed bed column, the CCTC process keeps the resin in the form of a slurry which flows through the static mixers and hollow fiber membrane modules in a countercurrent staged configuration to maximize the system performance. The static mixers provide residence time for interactions (e.g., binding of the mAb to the resin) while the hollow fiber membranes retain the large resin particles while permeating dissolved species like proteins and salts through the membrane. A schematic of the CCTC

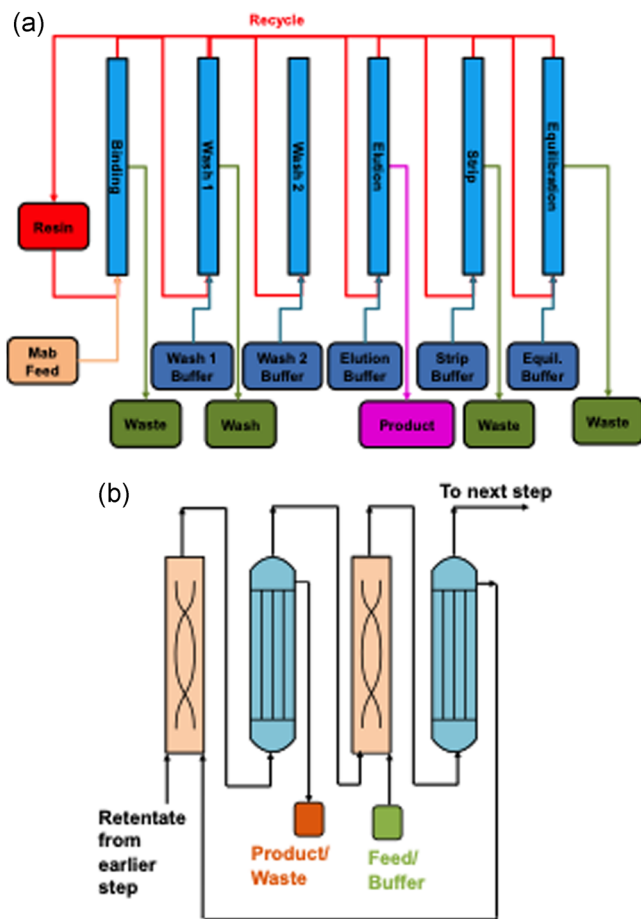


FIGURE 1 (a) Schematic of the CCTC system with continuous recycle of the protein A resin. The system consists of six steps: (1) binding, (2) wash-1, (3) wash-2, (4) elution, (5) strip, and (6) equilibration. (b) Schematic of a single step in the CCTC system showing two stages, each consisting of a static mixer and hollow fiber membrane module. CCTC, continuous countercurrent tangential chromatography [Color figure can be viewed at wileyonlinelibrary.com]

system is shown in Figure 1. The main design parameters for each step are the number of stages and the ratio of the permeate to retentate flow in that stage, both of which affect the product yield (in the elution step) and the degree of impurity removal (in the washing and strip steps).

The CCTC system has multiple advantages over column chromatography: (a) Operation of the CCTC system is continuous with steady-state product outlet concentration. Batch and multi-column systems (Godawat et al., 2012; Holzer, Osuna-Sanchez, & David, 2008; Muller-Späth et al., 2010) operate with continuous loading, but the outlet product concentration varies cyclically with time (peaks). (b) CCTC uses countercurrent staging, with the buffer and resin flowing in opposite directions to increase separation efficiency. (c) CCTC uses the resin as a slurry, significantly reducing the pressure drop in the CCTC system relative to packed columns (typically <15 psi), enabling the flow path to be manufactured with standard tubing and off-the-shelf disposable plastic materials (Dutta et al., 2017, 2015). This eliminates the need for any complex cleaning or sanitization cycles and is ideal for integration in single-use manufacturing facilities.

CCTC has been successfully used for initial capture of mAb from clarified cell culture fluid (CCF) using commercial protein A affinity resins, demonstrating productivities of up to 55 g of mAb/L of resin/hr, approximately fivefold increase over batch columns (Dutta et al., 2015). The CCTC system has also been used with the mixed-mode postcapture operation for mAb aggregate reduction. The mixed-mode operation of CCTC also showed higher productivity than batch column chromatography (10X) while maintaining similar yield and product quality. CCTC optimization showing the trade-off between yield, productivity, and buffer usage is discussed in detail by Dutta et al. (2016).

Another advantage of CCTC is that the pressure drop through the system is independent of particle size, in contrast to the $1/d_p^2$ dependence in column chromatography. Thus, CCTC could potentially be operated with much smaller particles than can be packed in columns, significantly reducing internal diffusional mass transfer resistance. In addition, the low-pressure operation of the CCTC makes it possible to use softer more highly porous resins, with enhanced dynamic binding capacity.

The objective of the current study was to obtain quantitative performance data for a CCTC system operating with a custom protein A affinity resin having a smaller particle size and greater porosity (lower cross-linking density). Experimental data for mAb capture demonstrate a greater than two-fold increase in productivity compared with results obtained using CCTC with standard (large particle diameter) resins, and a greater than 10-fold increase in productivity compared with conventional packed column (batch) chromatography.

2 | MATERIALS AND METHODS

2.1 | Materials

A harvested fed-batch CHO cell culture fluid at 2.3 g/L titer was prepared at Merck & Co., Inc., Kenilworth, NJ. The feedstock was stored at -80°C and then thawed and sterile filtered (Asepti-Cap®, Advanced Microdevices, India) before use. Hollow fiber membranes from Spectrum Labs (49–157 cm² membrane area, 0.65 μm pore size, 0.75 mm ID, and 41.5 cm length) and static mixers from Koflo Corporation (L = 28.6 cm, ID = 1.0 cm) were used to construct each stage in the CCTC system. Two different protein A capture resins were used: POROS MabCapture A from Thermo Fisher Scientific (Waltham, MA) and custom made prototype Praesto D5980 resin from Purolite Corporation (Bala Cynwyd, PA). The POROS MabCapture A has an average diameter of approximately 45 μm and is made of poly (styrene divinyl benzene). The Praesto D5980 resin is smaller (~25 μm) with an agarose backbone. MabCapture A resin will be referred to as Resin 1 while the Purolite resin will be referred to as Resin 2. All buffers were filtered through AseptiCap KS Polyethersulfone Capsule Filters with 0.45/0.2 μm pore size (Advanced Microdevices, India).

2.2 | Equilibrium binding capacity

The equilibrium binding capacity was determined by mixing 5 ml of 14% resin slurry with the mAb feedstock. The resulting mixture was stirred continuously for 15 min, with 1 ml samples of the fluid phase obtained through 0.22 μm syringe filters (polyethersulfone membrane, Advanced Microdevices, India). The concentration of free (unbound) antibody was evaluated by high performance liquid chromatograph (HPLC) using an analytical protein A column (Thermo Fisher Scientific). The mass of bound mAb was determined from the measured fluid phase concentrations using a simple mass balance.

2.3 | Batch uptake kinetics

The uptake (binding) kinetics for each resin were evaluated in batch. 10 ml of 14–17% resin slurry was mixed with a known volume of mAb feedstock. 1 ml samples of the fluid phase were collected at fixed time points (through 0.22 μm syringe filters) to evaluate the free mAb concentration as a function of time. The mass of bound mAb was determined from the measured fluid phase concentrations using a mass balance.

2.4 | CCTC experiment

A schematic of the CCTC system used for this study is shown in Figure 1. The process consisted of six chromatographic steps: (a) binding, (b) wash 1, (c) wash 2, (d) elution, (e) strip, and (f) equilibration. The wash-1 buffer was high salt with 10 mM sodium phosphate, 0.7 M NaCl, pH 6.6, while the wash-2 buffer was low salt with 10 mM sodium phosphate, pH 6.6. The elution was performed with 10–50 mM acetic acid for a target eluate pH of 3.7–4.2. The strip buffer was 0.18 M HCl, 0.18 M NaCl, 7 mM acetic acid, pH 2.1, while the equilibration buffer was 17 mM sodium phosphate, pH 7.8. Each CCTC step consisted of multiple stages with each stage composed of a static mixer and hollow fiber membrane. The ratio of permeate/retentate flow rates (γ) and the number of stages (N) were chosen to achieve >90% yield and at least 3-log impurity removal, discussed in more detail in later sections. The buffer and resin flow rates were controlled using peristaltic pumps (Watson Marlow, Wilmington, MA), with the pressures monitored continuously using Pendotech pressure sensors. The eluate permeate was monitored with an ultraviolet (UV)-detector at 280 nm manufactured by Pendotech; the flow-cell path length was set to stay within the linear range of detection. All buffer inlets and CCF were sterile filtered through 0.45/0.2 μm capsules (AseptiCap KS, Advanced Microdevices, India). Permeate samples from all steps were collected for off-line analysis of HCP, DNA, aggregate content, and titer.

2.5 | Single-pass tangential flow filtration

Single-pass tangential flow filtration (SPTFF) was performed with a Pall Cadence Inline Concentrator with 30 kD cellulose membrane having 0.065 m^2 membrane area (Pall Corp., Port Washington, NY CD030T010407). The module was integrated directly with the CCTC eluate permeate stream, giving a feed flux of 32 LMH. The retentate flux was set at 4 $\text{L m}^{-2} \text{h}^{-1}$ (LMH) by a peristaltic pump on the retentate exit line, providing an 8X concentration factor.

2.6 | Column purification

Batch chromatography experiments were performed at a constant linear velocity of 300 cm/h with a Millipore Vantage 100 $\times$ 11.3 mm column packed with Resin 1. The sequential chromatographic steps, including wash 1, wash 2, elution, and equilibration, were consistent with the CCTC experiments having the same buffering species and similar conductivity. The strip buffer was 100 mM acetic acid while the clean-in-place buffer was 100 mM sodium hydroxide. The batch elution pool volume corresponded to 4.5 column volumes.

2.7 | Assays

Residual host-cell DNA levels were determined to utilize an in-house probe-based quantitative polymerase chain reaction (qPCR) method. Isolated genomic DNA and primer/probe mixture were developed at Merck & Co., Inc., Kenilworth, NJ. DNA extraction and qPCR reaction used an AutoMate Express, a 7500 Fast Real-Time PCR System, and 96-well reaction plates from Applied Biosystems (Thermo Fisher Scientific).

mAb concentrations were determined using an Agilent 1100 HPLC equipped with a Poros A/20 column (1-5022-24) and a multi-wavelength detector set to 280 nm. Mobile phase A (MPA) was 1X Phosphate buffered-saline at pH 7.4; Mobile phase B (MPB) was 12 mM HCl with 150 mM NaCl. The column was operated at a constant flow rate of 3 ml/min. The column was first equilibrated with MPA, a 50 μL sample was injected, and the column was washed with MPA for 0.5 min. MPB was then used for 1 min followed by equilibration with MPA for 1.5 min. The peak area was used to calculate the mAb titer based on a comparison with a standard calibration curve.

Monoclonal antibody aggregates were quantified using a YMC-Pack Diol-200 300 $\times$ 8.0 mm column with 5 μm particle size (YMC Co., Ltd., Kyoto, Japan). The mobile phase was 50 mM sodium phosphate with 200 mM NaCl at pH 7.0. The column was operated for 30 min using an isocratic method at 0.5 ml/min flow rate. The UV280 response was calibrated against reference standards to qualify all measurements.

Residual host cell proteins were determined by enzyme-linked immunosorbent assay using a Chinese hamster ovary protein

Cygnus kit (P/N F550, Cygnus Technologies, Southport, NC) and a SpectroMax M5 reader. The procedure was automated with a Tecan Evo 200 replicating the manufacturer's protocol (Cygnus Technologies, 2013).

3 | RESULTS AND DISCUSSION

3.1 | Binding capacity and uptake kinetics

Three prototype Purolite resins of different particle sizes and pore structures were tested along with the 45 μm Poros resin (Resin 1) using a model (purified) mAb. The static binding capacity (Q_m) was evaluated from batch uptake experiments using appropriate mass balances:

$$Q_m = \frac{(C_0 - C_f)V_s}{\phi V_R} \quad (1)$$

$$V_s = V_R(1 - \phi) + 0.3V_R\phi + V_M \quad (2)$$

where C_0 is the initial mAb concentration after mixing the mAb solution with the resin slurry (but before any binding to the resin), C_f is the final concentration of mAb in solution (after binding equilibrium is reached), V_s is the total volume of the fluid phase in the final solution, V_R is the volume of the resin slurry, V_M is the volume of mAb feedstock, and ϕ is the resin concentration of the slurry (determined from the settled resin volume, vol/vol).

The equilibrium binding capacities for the three different Purolite prototypes were 51.2, 24.4, and 58.2 g/L, with the highest capacity obtained with the resin having a 3.5% agarose cross-linking density and a 25 μm particle size. Uptake kinetics for each resin was evaluated as described previously (Dutta et al., 2015), giving effective rate constants for the three Purolite prototypes of 0.19, 0.69, and 0.83 min^{-1} , respectively. The larger rate constant for the small particle size resin was a direct result of the reduction in diffusion mass transfer resistance. Based on these data, prototype D5980 (the third of the three resins listed above) was chosen for further study. Resin 2 in this paper corresponds to this Purolite resin prototype.

The chosen Purolite prototype (Resin 2) and 45 μm Poros (Resin 1) were used to determine the binding capacity for the mAb to be used in the CCTC experiments. The equilibrium binding capacities were 26 and 66 g/L for Resin 1 and Resin 2, respectively. The higher binding capacity for Resin 2 is likely due to the lower cross-linking density, which provides greater access to the binding sites within the particle, increasing the utilization of the higher ligand density in Resin 2.

3.2 | CCTC run—design of the binding step

The operating conditions for the CCTC system were chosen based on previously developed equations and protocols. The ratio of the mAb

feed flow rate to the resin slurry flow rate in the binding step (γ_b) was chosen according to the following equation:

$$\gamma_b = \frac{LQ_m\phi}{C_m} \quad (3)$$

where L is the % loading and C_m is the mAb concentration in the feed. Equation (3) gives γ_b for Resin 1 of 1.4 while that for Resin 2 was 2.0. The γ_b for Resin 2 was only 50% higher even though the binding capacity was more than 100% greater than the binding capacity of Resin 1. Reducing the loading on Resin 2 reduced the required binding time which maximized the resin productivity as determined by the CCTC model (Dutta et al., 2016). Even at that lower % loading, the absolute loading (35 g of mAb/L of resin) for Resin 2 was significantly higher compared with Resin 1 (21 g of mAb/L of resin).

The binding step in the CCTC system was sized for the necessary residence time as measured by batch uptake kinetics experiments as shown in Figure 2. The experiments for the two resins were performed with different ratios of mAb to the resin to simulate the behavior in the CCTC process based on the different values of γ_b . The uptake kinetics for Resin 2 was faster than those for Resin 1, with the time required for the mAb concentration to decrease to 0.095 g/L equal to 8 min for Resin 2 compared with >12 min for Resin 1. The design of the binding step based on the uptake kinetics data is described in our previous publication (Dutta et al., 2015). In both cases, a two-stage binding step was used with an "after binder"

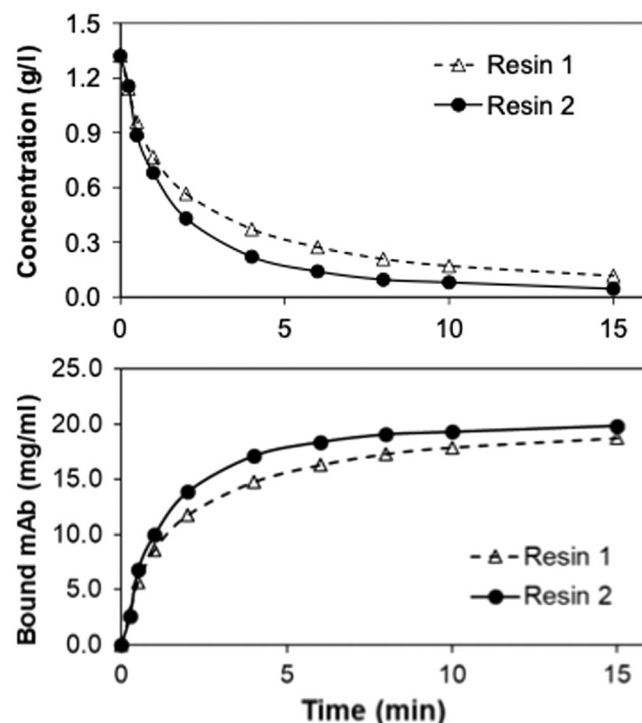


FIGURE 2 Comparison of binding kinetics for the two resins. Resin 2 (Praesto D5980) with its smaller particle size showed faster kinetics than Resin 1 (MabCapture A) because of lower diffusional mass transfer resistance. The lines are to guide the eye

(a small static mixer placed after the second hollow fiber membrane module to increase binding of the mAb before the washing stage). For Resin 1, the residence times were 5, 4, and 6 min for the first stage, second stage, and after-binder, respectively. The corresponding times for Resin 2 were 1.9, 3.4, and 2.6 min. Thus, the total residence time for Resin 2 (7.9 min) was approximately one-half that for Resin 1 (15 min).

3.3 | Design of nonbinding steps

A detailed discussion of the design of CCTC systems is presented in Dutta et al. (2016). The two important parameters for each step are the number of stages and the ratio of the permeate/retentate flow (γ) in that stage, both of which affect the product yield (in the elution step) and the degree of impurity removal (in the washing and strip steps). It has been shown in previous publications (Dutta et al., 2017) that the critical flux is usually $>650 \text{ L}\cdot\text{m}^{-2}\cdot\text{hr}^{-1}$ (LMH) for the typical operating conditions used in the CCTC system; this was also true for the smaller Resin 2. The required membrane surface area for each hollow fiber module was chosen so that the permeate flux was $<50\%$ of the critical flux (giving an upper limit of $325 \text{ L}\cdot\text{m}^{-2}\cdot\text{hr}^{-1}$).

3.4 | Optimizing the design of the CCTC system for 45 μm resin (Resin 1)

The first run (Run 1A) for Resin 1 was conducted according to the conditions given in Table 1 with $\gamma_0 = 1.4$. The wash-1 and wash-2 steps were designed with two and three stages, respectively, which should provide at least 99% impurity removal in each wash. The elution step was designed with four stages to achieve a 98% theoretical mAb yield. The results from the run are summarized in Table 2. The overall mAb yield for the CCTC process was 97% with a productivity of 33 g of mAb/L of resin/hr, a 3X improvement over the batch column. The buffer consumption for this run was just under 2 L/g of mAb. The process provided greater than a 2-log reduction in HCP and DNA, and the monomer content in the eluted product was 98.8%.

TABLE 1 Design parameters for different CCTC steps

Step	No. of stages	γ	J_{Actual} ($\text{L}\cdot\text{m}^{-2}\cdot\text{hr}^{-1}$)
Binding	2	1.4–2.0	115–148
Wash 1	2–3*	2.25–3.60	235–299
Wash 2	3–4*	2.00–3.00	256–294
Elution	4	2.30	105–169
Stripping	2	2.25	235
Equilibration	4	2.00	294

Abbreviation: CCTC, continuous countercurrent tangential chromatography.
*Conditions for modified CCTC run to reduce buffer consumption.

TABLE 2 Yield, productivity, and product quality comparison of CCTC-Resin 1, CCTC-Resin 2, and batch column-Resin 1

Performance	CCTC-Resin 1 Run 1	CCTC-Resin 1 Run 2	CCTC-Resin 2 Run 1	Column
Yield (%)	97	91	91–95	95–99
Resin productivity (g/L/hr)	34	38	140	12
HCP (ppm)	7025	245	420	245
DNA (ppm)	10.0	0.2	<LOQ	<LOQ
Monomer in product	98.8	99.6	99.5	>99

Notes: The HCP in the feed was 1,062,000 ppm; the leached protein A was not analyzed in this study; only Resin 1 data for column is available; LOQ indicates lower than detection limit.

Abbreviations: CCTC, continuous countercurrent tangential chromatography; HCP, host cell protein; LOQ, limit of quantification.

In Run 1B, the CCTC impurity removal was optimized by increasing the number of stages in wash-1 (to $N = 3$) and wash-2 (to $N = 4$) and by increasing the ratio of the buffer to resin flow rate in the wash steps as shown in Table 1. HCP removal under these conditions increased to 3.4-log while DNA removal was 4-logs (Table 2).

3.5 | Comparison of CCTC system performances for 45 and 25 μm resins

The CCTC system was run with both Resin 1 and Resin 2 with results summarized in Table 2 along with data for a conventional packed column using the larger particle size resin (Resin 1). The maximum steady-state productivities of the CCTC system were 38 and $140 \text{ g}\cdot\text{L resin}^{-1}\cdot\text{hr}^{-1}$ for Resin 1 and Resin 2, respectively. The smaller particle has faster update kinetics, reducing the static mixer volumes (i.e. smaller residence times) required in the binding stage, and increasing the dynamic binding capacity; both factors contributed to increased system productivity. The productivity of the CCTC system with Resin 2 was more than 10 times higher than the packed column ($12 \text{ g}\cdot\text{L resin}^{-1}\cdot\text{hr}^{-1}$). The eluate mAb concentration for Resin 2 was 1.9 g/L (70% higher than with Resin 1).

3.6 | Stability of the CCTC operation

Both Resin 1 and Resin 2 were used for the purification of mAb using the CCTC system. Typical results for the mAb concentrations in the eluted product stream during a 3-hr CCTC run with Resin 2 (Run 2) and 5.5 hr run with Resin 1 (Run 1B) are shown in Figure 3. The A_{280} values remained essentially constant throughout the CCTC operation, in contrast to the periodic elution peaks obtained in multi-column chromatography systems. The various changes in the A_{280} profile for Resin 1 were due to changes in

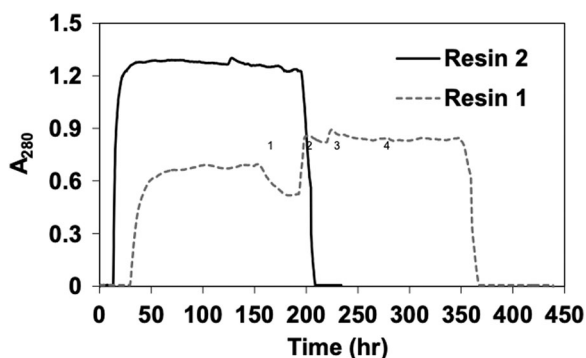


FIGURE 3 A_{280} signal for the CCTC run using Resin 1 and Resin 2. The pressure profile for Resin 2 was stable throughout the run. The pressure profile for Resin 1 jumped after 150 min due to intentional changes in elution conditions during the run to study the effect of pH and γ on mAb yield. (1) $\gamma = 2.3$, pH 4.2, 74% Yield; (2) $\gamma = 3.0$, pH 4.0, 82% yield; (3) $\gamma = 2.3$, pH 3.9, 91% yield; (4) $\gamma = 2.3$, pH 3.7, 91% yield. CCTC, continuous countercurrent tangential chromatography; mAb, monoclonal antibody

operating conditions for on-line optimization testing, something that is easily done in CCTC (in contrast to batch columns). In this case, a decrease in elution pH (from 4.1 to 3.7) at a constant gamma increased the product yield leading to an increase in the UV signal. An increase in gamma for the elution step led to a decrease in pH of the eluted product stream, leading to higher product yield, however, the UV signal slightly decreased due to increased dilution of the product. These “in-run” modifications demonstrate the fast responsiveness of CCTC to process changes, in turn facilitating simpler integration of this platform with appropriate process analytical technology (PAT) and advanced process control strategies.

The pressure profiles during CCTC operation were stable and <10 psi for all steps. The pressure profiles for Resin 2 are shown in Figure 4. The pressure profiles were also stable after the values of γ for the wash-1 and wash-2 steps were increased to achieve higher impurity removal. This demonstrates the ability to adjust/change process parameters during the run while keeping the hydrodynamics of the CCTC system robust and linear.

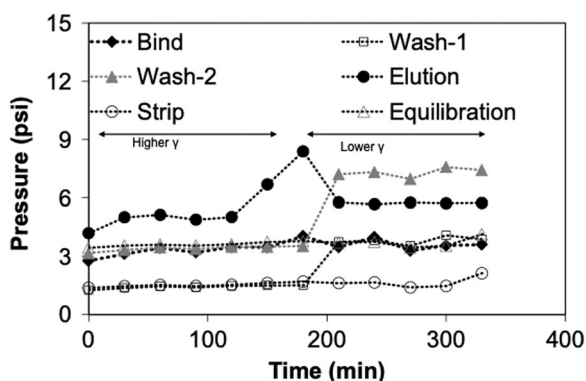


FIGURE 4 Pressures in different steps during CCTC run using Resin 2. CCTC, continuous countercurrent tangential chromatography

3.7 | Reduction in buffer consumption and an increase in product concentration

Two challenges associated with CCTC are lower product concentration and higher buffer usage compared with batch column operation. The product concentration obtained using the smaller particle size, higher binding capacity Resin 2 increased the eluate titer twofold compared with that achieved using Resin 1. Similarly, buffer consumption with Resin 2 was decreased to 1.3 L/g of mAb compared with 2.3 L/g of mAb for Resin 1 for similar product quality. To decrease buffer consumption further, the number of stages was increased, the resin concentration was increased, and γ_b was increased while γ in some steps was decreased. The effect of staging, resin concentration, γ_b , and γ on buffer consumption and impurity removal are discussed in more detail in our previous article (Dutta et al., 2016). The operating conditions used for this optimized run (Run 3) are detailed in Table 3. To increase the final product concentration, an SPTFF was integrated with the CCTC system directly on the output from the elution step. The use of SPTFF in this run resulted in a product stream with a steady-state concentration of 16 g/L, more than a 10-fold increase compared with previous eluate titers. Moreover, the product quality was slightly improved over previous runs, with a 4.0-log DNA reduction and 3.6-log HCP reduction. The aggregate content of the eluted product showed only 0.5% HMW as measured by HPLC-SEC.

Another experiment with Resin 2 was performed to optimize the system for higher yield. This experiment achieved 95% yield (vs. 91% in the previous run). Unfortunately due to material limitations, the overall flow rates were not optimal and in this case, the system achieved 70 g·L⁻¹·hr⁻¹ productivity which was still 5X more than the batch column.

3.8 | Scale-up of the CCTC system and comparison with a batch column

Model calculations were performed for the purification of CCF produced from a 2000 L batch bioreactor using CCTC and batch

TABLE 3 Design parameters for different CCTC steps for Run 3 with Resin 2

Step	No. of stages	γ
Binding	2	2.92
Wash 1	4	3.00
Wash 2	4	2.40
Elution	4	2.30
Stripping	3	2.25
Equilibration	4	2.00

Abbreviation: CCTC, continuous countercurrent tangential chromatography.

TABLE 4 Scale-up comparison between CCTC and column operations for 9-hr process time

	Column: 45 μm	CCTC: 45 μm	CCTC: 25 μm
Total operation time (hr)	9.9	8.5	8.5
Cycle time (hr)	3.0	0.5	0.25
Resin volume (L)	41.7	17.6	4.2
Total buffer volume (L)* (Buffer concentrates in CCTC)	4,200	2,500	1,200
WFI volume (L)**	NA	12,200	5,600
Total water volume (L)***	4,200	14,700	6,800
CCTC in-line mixing			
Average buffer tank volumes (L)****	700	450	200

Abbreviations: CCTC, continuous countercurrent tangential chromatography; WFI, water for injection.

*Using 10X buffer concentrate for CCTC in wash-2, elution, and equilibration steps; 3X concentrate in strip.

**WFI is assumed to not require additional tankage.

***Total water volume is the sum of WFI and buffer volume.

****Average buffer tank volumes are lower with CCTC because of steady-state in-line mixing.

columns based on the results with Resin 1 (Run 1B) and Resin 2 (Run 2). The results for the batch column packed with 45 μm particles and the CCTC system with both the 45 and 25 μm particles are summarized in Table 4. Calculations were not performed for the batch column using the smaller particle size since the pressure drop in the column using 25 μm particles would be prohibitive. The cycle time in the CCTC system is significantly shorter than in the batch column (15 min for Resin 2 vs. 3 hr for column). Extension of run time for Resin 2 in the CCTC system to 24 hr would give 96 cycles for the protein A resin, enabling full lifetime utilization of this resin during a single run. The resin volume required with the 25 μm resin in the CCTC system is 3X less than CCTC with the 45 μm resin and >10X less than batch column requirements. This shows very significant potential for cost savings. The resin volume requirement for the column system could be reduced by using a multi-column arrangement (e.g., periodic countercurrent chromatography), although resin savings are typically less than a factor of two (Godawat et al., 2012).

Although the CCTC system uses more buffer than a packed column, one important advantage of CCTC is the simplicity of using in-line buffer dilution due to the simultaneous and independent steady-state buffer feeding of all steps. As shown in Table 4, the CCTC system with the 25 μm resin would require 40% higher total water volume compared with the batch column. However, the simplicity of inline CCTC buffer dilution would reduce buffer tankage and floor space by over 70% when compared with conventional batch column operation (200 L average tank size for CCTC vs. 700 L for batch column).

4 | CONCLUSIONS

The results from this study demonstrate the significant improvements achieved using a custom smaller particle size resin that takes advantage of the wider design space offered by the format of the CCTC platform. Because the pressure drop in the CCTC system is independent of particle size, a smaller particle size resin with higher binding capacity due to lesser cross-linking and faster kinetics significantly reduced the volume of the protein A resin required. The combination of improved uptake kinetics and greater dynamic binding capacity provided a 4X reduction in resin volume and nearly a 3X increase in productivity up to 140 g of mAb·L resin⁻¹·hr⁻¹, which is more than 10X higher than in a conventional packed column.

Further optimization of the CCTC system was achieved by changing the number of stages and ratio of buffer or resin flow rate to enhance impurity removal and buffer utilization. A CCTC system with two wash-1 and three wash-2 stages with $\gamma = 2.25$ provided 2.2-log reduction for both HCP and DNA. This was increased to 3.4-log reduction for HCP and 4.0-log reduction for DNA by increasing the number of wash-1 stages to three, the number of wash-2 stages to four, and increasing γ to 3.6. Buffer consumption of the CCTC was reduced by 54% by using the small particle size resin, although the CCTC system still uses greater total water volume and buffer components than a comparable batch column. However, CCTC enables simple steady-state in-line buffer dilution which significantly reduces the required buffer tankage by 70% versus a traditional batch column. The final product concentration was increased by direct integration of SPTFF with the elution step, providing a reduction in the volume of the product stream to facilitate the design and operation of subsequent downstream processing steps. This would enable the direct coupling of CCTC within a highly integrated biomanufacturing process forming the basis for a new continuous processing platform. Although current SPTFF systems are quite expensive, recent work has demonstrated the potential of using a low-cost hollow fiber module for SPTFF, which would also facilitate integration in a fully single-use process (Yehl & Zydney, 2019).

ACKNOWLEDGMENTS

We thank Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Kenilworth, NJ, for contributing the mAb feedstock and analytical support as well as for sponsoring these studies. We also thank the Purolite Corporation for producing the small particle size agarose resin used in these studies.

CONFLICT OF INTERESTS

Several of the authors (DF, AKD, JT, ALZ, and OS) have financial interests in Chromatan Corporation, which is directly involved in the commercialization of the continuous countercurrent tangential chromatography system.

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How to cite this article: Fedorenko D, Dutta AK, Tan J, et al. Improved protein A resin for antibody capture in a continuous countercurrent tangential chromatography system. *Biotechnology and Bioengineering*. 2020;117:646–653. <https://doi.org/10.1002/bit.27232>